

AERO-GUARD MISSION CONTROL AUDIT LEDGER

Autonomous Spacecraft Health Management (FDIR) • Closed-Loop Telemetry & Operator Audit Trail

MET STATUS: ACTIVE CRUISE
ORBIT: LEO 552 KM • INC 97.6°
HARDWARE: HITL PYSERIAL BRIDGE

LOG BUFFER: 54 / 10000 LINES (FIFO ROLLING)

1. EXECUTIVE AUDIT METRICS & BUFFER TELEMETRY

TOTAL LOGS RECORDED	OPERATOR ACTIONS	EARLY PREDICTIONS	ANOMALIES DETECTED	RL MITIGATIONS	RECOVERIES
54	3	14	1	1	1
Capacity: 0.5%	Manual & Gates	Precursor Warnings	Multi-Class FDIR	Q-Learning Policy	Nominal Restored

2. SPACECRAFT SUBSYSTEM HEALTH & 5-FAULT COMPLIANCE ARCHITECTURE

Subsystem Component	Target Spacecraft Fault	Detection Precursor / Trigger	Closed-Loop Mitigation Policy	Recovery Verification Standard
1. PCDU Bus-A Power Distribution	Internal Micro-Short (Dendrite penetration)	Current > 4.5A spike Bus Voltage < 2.60V	SAFETY_OVERRIDE / SAFE_MODE Latching relay trips (<20ms); cross-straps to Bus-B	NASA-HDBK-4008 \$4.2 OBC isolated at 9.2W
2. Li-ion 8S4P Pack Secondary Energy	Thermal Runaway (Joule heating accumulation)	dT/dt > 1.2°C/s Cell Temp T > 50°C	LOAD_SHEDDING (-35% Payload) RPC trips; BCR to Trickle; ADCS slews +18.5° to 3K sink	ECSS-E-ST-20C Joule heat drops 85%
3. BCR & Shunt Reg Solar Ingestion	Deep Undervoltage (Extended eclipse excursion)	Voltage < 2.90V SOC < 15%	LOAD_SHEDDING (Tier-2 UVLS) -75% Load shed; ADCS B-dot detumble slews arrays to Sun	ISRO-URSC-PCDU-E PS-04 Bus restores >3.3V
4. Li-ion Electrodes Electrochemical Cells	High Impedance / Aging (SEI Layer thickening)	R_int > 0.120 Ω Transient ESR ripple	HIGH_SENSITIVITY_PREARM RL lowers Tau=0.35; caps max discharge rate to 0.4C (1.5A)	NASA-HDBK-4008 Voltage sag stabilized
5. Sensor ADC Transducer Telemetry Bus	Sensor Glitch / Noise (Radiation SET transient)	> 3-sigma Z-Score outlier 12Hz ADC ringing	AI_RAG_REASONER + EWMA EWMA rejects spike; AI raises Tau=0.70 to prevent false trip	False Alarm Rate < 1.2% Payload stays online

3. CHRONOLOGICAL MISSION AUDIT LEDGER (UP TO 10,000-LINE BUFFER)

Full audit trail of all frontend user actions, streaming telemetry, early precursor predictions, anomaly triggers, RL mitigation actuations, and recovery events:

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#54 T+00:01:02	INFO	TELEMETRY FRAME	EPS CORE	V: 4.11V • I: 1.79A • T: 23.3°C • SOC: 89% • P_ens: 0.00	Telemetry Cruise: V=4.107V I=1.787A T=23.3°C SOC=88.6% P_ens=0.001
#53 T+00:00:59	INFO	TELEMETRY FRAME	EPS CORE	V: 4.09V • I: 1.74A • T: 23.5°C • SOC: 89% • P_ens: 0.00	Telemetry Cruise: V=4.093V I=1.741A T=23.5°C SOC=88.5% P_ens=0.000
#52 T+00:00:59	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.15V • I: 1.75A • T: 24.2°C • SOC: 89% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.000 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#51 T+00:00:57	INFO	TELEMETRY FRAME	EPS CORE	V: 4.07V • I: 1.79A • T: 23.2°C • SOC: 89% • P_ens: 0.00	Telemetry Cruise: V=4.074V I=1.792A T=23.2°C SOC=88.5% P_ens=0.001
#50 T+00:00:54	INFO	TELEMETRY FRAME	EPS CORE	V: 4.16V • I: 1.81A • T: 23.8°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.163V I=1.811A T=23.8°C SOC=88.5% P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#49 T+00:00:54	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.16V • I: 1.81A • T: 23.8°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#48 T+00:00:51	INFO	TELEMETRY FRAME	EPS CORE	V: 4.09V • I: 1.78A • T: 23.5°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.093V I=1.776A T=23.5°C SOC=88.4% P_ens=0.001
#47 T+00:00:50	ACTION	OPERATOR ACTION	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: OPEN_MISSION_AUDIT_CONSOLE. Opened 10k FIFO Mission Audit Ledger
#46 T+00:00:49	INFO	TELEMETRY FRAME	EPS CORE	V: 4.10V • I: 1.87A • T: 23.3°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.101V I=1.870A T=23.3°C SOC=88.4% P_ens=0.001
#45 T+00:00:46	INFO	TELEMETRY FRAME	EPS CORE	V: 4.14V • I: 1.88A • T: 23.0°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.141V I=1.883A T=23.0°C SOC=88.4% P_ens=0.001
#44 T+00:00:45	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.11V • I: 1.86A • T: 23.5°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#43 T+00:00:44	INFO	TELEMETRY FRAME	EPS CORE	V: 4.10V • I: 1.81A • T: 23.1°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.096V I=1.808A T=23.1°C SOC=88.4% P_ens=0.001
#42 T+00:00:43	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.11V • I: 1.74A • T: 23.4°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#41 T+00:00:41	INFO	TELEMETRY FRAME	EPS CORE	V: 4.12V • I: 1.75A • T: 23.1°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.123V I=1.755A T=23.1°C SOC=88.3% P_ens=0.001
#40 T+00:00:41	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.08V • I: 1.91A • T: 23.2°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#39 T+00:00:39	INFO	TELEMETRY FRAME	EPS CORE	V: 4.10V • I: 1.78A • T: 23.4°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.098V I=1.776A T=23.4°C SOC=88.3% P_ens=0.001
#38 T+00:00:38	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.07V • I: 1.79A • T: 22.4°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#37 T+00:00:36	INFO	TELEMETRY FRAME	EPS CORE	V: 4.05V • I: 1.75A • T: 22.9°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.046V I=1.750A T=22.9°C SOC=88.3% P_ens=0.001
#36 T+00:00:35	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.11V • I: 1.75A • T: 23.0°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#35 T+00:00:34	INFO	TELEMETRY FRAME	EPS CORE	V: 4.09V • I: 1.75A • T: 22.3°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.090V I=1.754A T=22.3°C SOC=88.3% P_ens=0.001
#34 T+00:00:31	INFO	TELEMETRY FRAME	EPS CORE	V: 4.14V • I: 1.80A • T: 22.6°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.138V I=1.801A T=22.6°C SOC=88.2% P_ens=0.001
#33 T+00:00:30	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.08V • I: 1.81A • T: 22.2°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#32 T+00:00:29	INFO	TELEMETRY FRAME	EPS CORE	V: 4.07V • I: 1.79A • T: 22.5°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.074V I=1.790A T=22.5°C SOC=88.2% P_ens=0.001

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#31 T+00:00:28	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.12V • I: 1.78A • T: 22.3°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#30 T+00:00:27	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.07V • I: 1.73A • T: 22.7°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.000 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#29 T+00:00:26	INFO	TELEMETRY FRAME	EPS CORE	V: 4.06V • I: 1.80A • T: 22.1°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.056V I=1.796A T=22.1°C SOC=88.2% P_ens=0.001
#28 T+00:00:24	INFO	TELEMETRY FRAME	EPS CORE	V: 4.09V • I: 1.77A • T: 22.4°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.089V I=1.767A T=22.4°C SOC=88.2% P_ens=0.001
#27 T+00:00:21	INFO	TELEMETRY FRAME	EPS CORE	V: 4.07V • I: 1.79A • T: 22.1°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.066V I=1.790A T=22.1°C SOC=88.1% P_ens=0.001
#26 T+00:00:21	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.07V • I: 1.79A • T: 22.1°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#25 T+00:00:19	INFO	TELEMETRY FRAME	EPS CORE	V: 4.07V • I: 1.87A • T: 22.6°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.074V I=1.875A T=22.6°C SOC=88.1% P_ens=0.001
#24 T+00:00:16	INFO	TELEMETRY FRAME	EPS CORE	V: 4.12V • I: 1.81A • T: 22.3°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.118V I=1.811A T=22.3°C SOC=88.1% P_ens=0.001
#23 T+00:00:14	INFO	TELEMETRY FRAME	EPS CORE	V: 4.10V • I: 2.51A • T: 22.5°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.098V I=2.509A T=22.5°C SOC=88.0% P_ens=0.001
#22 T+00:00:14	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.10V • I: 2.51A • T: 22.5°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#21 T+00:00:11	INFO	TELEMETRY FRAME	EPS CORE	V: 4.11V • I: 2.52A • T: 22.1°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.113V I=2.516A T=22.1°C SOC=88.0% P_ens=0.001
#20 T+00:00:09	INFO	TELEMETRY FRAME	EPS CORE	V: 4.10V • I: 2.51A • T: 22.3°C • SOC: 88% • P_ens: 0.00	Telemetry Cruise: V=4.104V I=2.507A T=22.3°C SOC=88.0% P_ens=0.001
#19 T+00:00:08	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.03V • I: 2.60A • T: 21.6°C • SOC: 88% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#18 T+00:00:06	INFO	SYSTEM	HITL PIPELINE	N/A	AERO-GUARD Master Pipeline initialized. Synthetic telemetry baseline loaded.
#17 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.71V • I: 2.10A • T: 23.2°C • SOC: 84% • P_ens: 0.01	Post-Recovery Cruise: Bus Voltage=3.71V, Load Current=2.10A, Core Temp=23.2C, SOC=84.5%
#16 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.71V • I: 2.10A • T: 23.2°C • SOC: 84% • P_ens: 0.01	Post-Recovery Cruise: Bus Voltage=3.71V, Load Current=2.10A, Core Temp=23.2C, SOC=84.5%
#15 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.71V • I: 2.10A • T: 23.2°C • SOC: 84% • P_ens: 0.01	Post-Recovery Cruise: Bus Voltage=3.71V, Load Current=2.10A, Core Temp=23.2C, SOC=84.5%
#14 T+00:00:00	RECOVER	RECOVERY EVENT	SPACECRAFT HEALTH	N/A	Nominal State Restored: LOAD_SHEDDING successfully mitigated THERMAL_RUNAWAY. Core temperature stabilized at 24.1C. Joule heating reduced by 85%. Nominal flight limits verified.
#13 T+00:00:00	SAFETY	SAFETY OVERRIDE	SAFETY GATEWAY	N/A	Latching relay pre-armed. Automatic load shedding executed via solid-state PCDU RPC-1/2 switch.

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#12 T+00:00:00	RL-POLICY	RL MITIGATION	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [THERMAL_RUNAWAY]. Projected 60s Temp: 39.8°C, Safety Score: 0.96
#11 T+00:00:00	INFO	AI RAG REASONING	AI AGENT RAG	N/A	[AI Diagnostic]: Subsystem entered High Risk critical envelope. Joule heating (I^2R) exceeded radiator cooling threshold. Citation: NASA-HDBK-4008: Thermal runaway mitigation protocol §4.2.1
#10 T+00:00:00	CRITICAL	FAULT ANOMALY	EPS CORE	V: 3.52V • I: 4.85A • T: 51.2°C • SOC: 79% • P_ens: 0.97	Anomaly Triggered: THERMAL_RUNAWAY Severity: CRITICAL. Core Temperature reached 51.2C with dT/dt > 1.8C/s. Multi-Class FDIR confirmed Thermal Runaway (98.4% conf).
#9 T+00:00:00	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 3.65V • I: 4.10A • T: 36.5°C • SOC: 82% • P_ens: 0.42	Early Precursor Detected: Thermal Gradient Surge (+1.4C/s) P(Anomaly)=0.420 >= Tau=0.70. Recommended Pre-Arm: LOAD_SHEDDING (-35% Payload)
#8 T+00:00:00	ACTION	OPERATOR ACTION	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_THERMAL_RUNAWAY. Operator injected Thermal Runaway fault bus test for 25.0s.
#7 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.72V • I: 2.45A • T: 22.4°C • SOC: 85% • P_ens: 0.01	Nominal Cruise: Bus Voltage=3.72V, Load Current=2.45A, Core Temp=22.4C, SOC=85.0%
#6 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.72V • I: 2.45A • T: 22.4°C • SOC: 85% • P_ens: 0.01	Nominal Cruise: Bus Voltage=3.72V, Load Current=2.45A, Core Temp=22.4C, SOC=85.0%
#5 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.72V • I: 2.45A • T: 22.4°C • SOC: 85% • P_ens: 0.01	Nominal Cruise: Bus Voltage=3.72V, Load Current=2.45A, Core Temp=22.4C, SOC=85.0%
#4 T+00:00:00	INFO	TELEMETRY FRAME	EPS CORE	V: 3.72V • I: 2.45A • T: 22.4°C • SOC: 85% • P_ens: 0.01	Nominal Cruise: Bus Voltage=3.72V, Load Current=2.45A, Core Temp=22.4C, SOC=85.0%
#3 T+00:00:00	ACTION	OPERATOR ACTION	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: SET_CAMERA_VIEW_IR_THERMAL. Operator switched telemetry 3D view to IR Thermal mode.
#2 T+00:00:00	INFO	SYSTEM	EPS CORE	N/A	AERO-GUARD Satellite Health Monitoring System initialized in LEO orbit 552km.
#1 T+00:00:00	INFO	SYSTEM	LOG ENGINE	N/A	Mission Audit Log Buffer cleared by operator command.